

Introduction to GAPS

Lecture 01: General Agent Problem Solving

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Panaversity Certified AI Development Program

Speaker Notes: Welcome the students to the Certified AI Development Program. Introduce the faculty and explain that today we move from prompt engineering to building autonomous agents.

| Learning Objectives

- ✓ Master the core evolution from static automation to Agentic AI.
- ✓ Understand the structural blueprint of the GAPS framework.
- ✓ Analyze the pillars of Perception, Reasoning, and Action.
- ✓ Explore the mechanics of self-correcting and self-healing systems.
- ✓ Learn to implement Human-in-the-Loop (HITL) for enterprise safety.

| Meet the Faculty & Leadership

Sir Zia Khan: Lead visionary with decades of global tech experience, spearheading the Pakistani AI revolution.

Waniya Kazmi: Software Engineer and AI Architect, focused on the practical implementation of Agentic systems.

Sir Ammar: Technical lead ensuring student success and structural mastery of AI infrastructure.

Our goal is to build world-class AI talent capable of leading the global tech landscape.



Program Vision: From Users to Builders

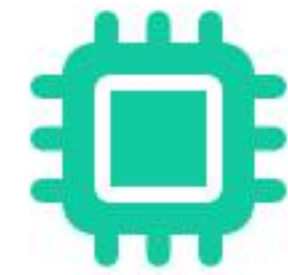
Speaker Notes: Transition into the core vision of Panaversity—moving away from just using tools to actually architecting them.

| Core Philosophies of the Program



Entrepreneurial

Focusing on building startups and product initiatives, not just employee skills.



Agentic First

Moving beyond prompts to building autonomous, multi-step agent architectures.



Enterprise Grade

Developing solutions that are scalable, secure, and production-ready for global markets.

| The Microwave Analogy

Traditional Appliance: Follows a fixed timer set by a human who acts as the primary sensor.

Smart AI Appliance:

- Uses internal sensors to scan food state.
- Reasons whether the food is frozen or at room temp.
- Calculates and adapts the heating time automatically.



The Evolution of AI Paradigms

Phase	Characteristics	Human Role
Traditional Automation	Fixed rules, rigid code, fails on edge cases.	Programmer/Operator
Generative AI	Single-turn output, creative prompting.	Continuous Prompter
Agentic AI	Autonomous loops, tool use, goal-oriented.	Goal Setter/Reviewer

Speaker Notes: Explain how we've moved through these phases. We are currently transitioning from the "Prompter" era into the "Goal Setter" era with AI Agents.

| Traditional Automation Limits

0%

Lack of Adaptability

Traditional rule-based systems have zero ability to handle unexpected scenarios. They follow a script and fail immediately when reality deviates from the hard-coded logic.

Defining Agentic AI

| Characteristics of AI Agents



Goal-Oriented

Driven by high-level objectives rather than granular instructions.



Tool Enabled

Able to use external tools like Python, SQL, or web search.



Self-Healing

Detects its own errors and re-plans to achieve the goal.

| The GAPS Framework Anatomy

General Agent Problem Solving (GAPS) is our core architectural model. It consists of three primary layers:

- **Perception (Sensors):** Ingesting data from the environment.
- **Brain (Reasoning):** The LLM core that plans and prioritizes.
- **Action (Effectors):** The tools that execute the plan.

 Digital neural network brain visualization

| Pillar 1: Perception (Sensors)

-  **Document Parsing:** Reading and understanding flat files (PDFs, CSVs).
-  **Database Monitoring:** Ingesting changes from SQL or Vector DBs.
-  **API Consumption:** Real-time data feeds from external services.
-  **Environment Observation:** Scanning logs and system metrics.

| Pillar 2: Brain (Reasoning)

The reasoning core is powered by high-capacity LLMs (like GPT-4o, Claude 3.5, or Gemini 1.5 Pro).

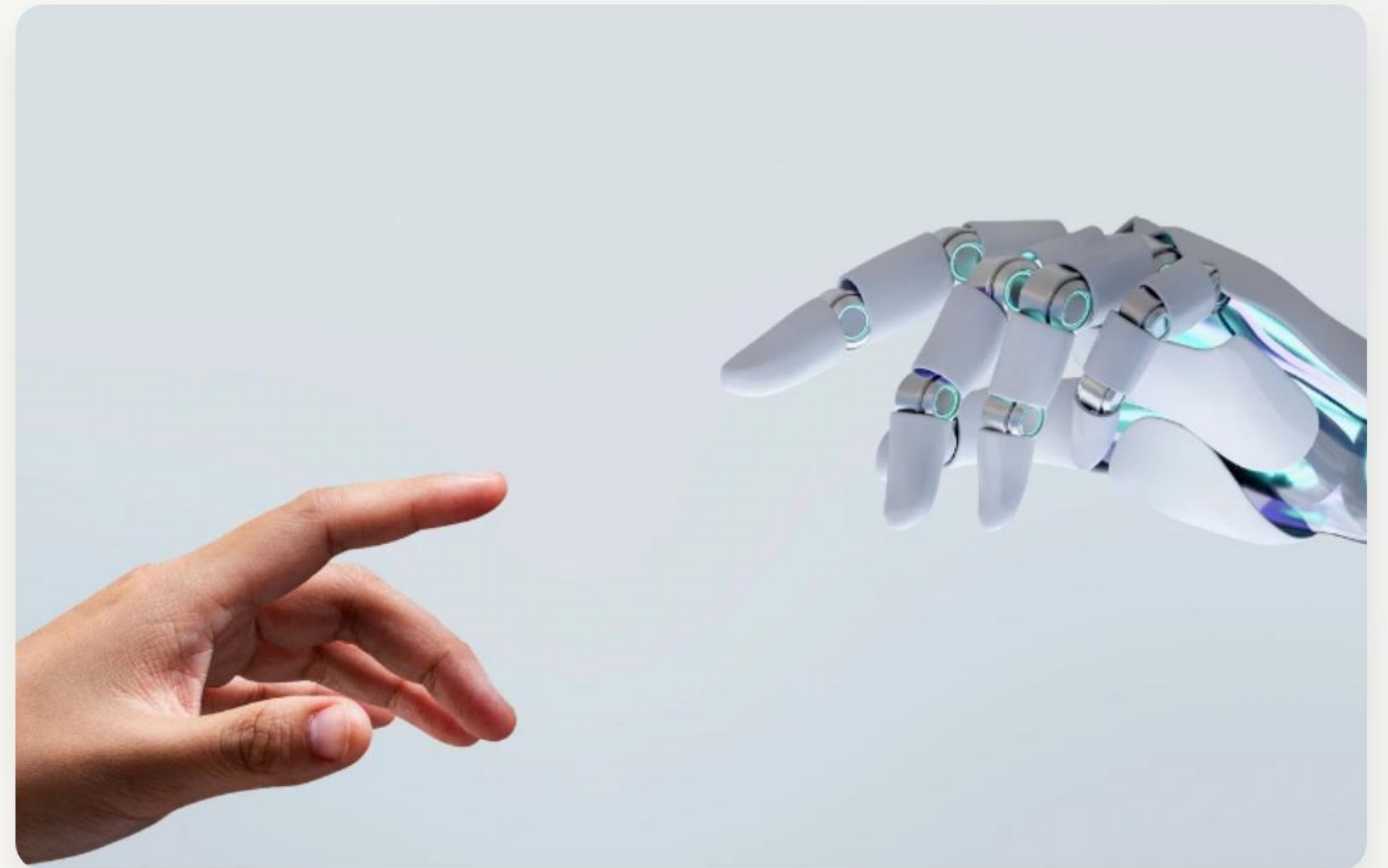
- **Goal Decomposition:** Breaking big tasks into small ones.
- **Memory Management:** Using context to keep track of progress.
- **Decision Logic:** Choosing the right tool for the current step.

 Glowing digital connections in a neural brain

| Pillar 3: Action (Effectors)

This is where the theoretical plan becomes reality through digital tools:

- **Code Execution:** Writing and running Python scripts locally.
- **API Mutation:** Sending requests to change states in external systems.
- **File Generation:** Creating reports, summaries, or new datasets.
- **Communication:** Messaging human users for approval or updates.



The Mechanics of Autonomy

Speaker Notes: Let's look at the process behind the agent's work cycle.

| The Self-Correction Loop

- ▶ **Step 1:** Agent executes a tool (e.g., Python code).
- 🐛 **Step 2:** Error occurs or task fails (runtime crash).
- 🔍 **Step 3:** Agent observes the error stack trace.
- ✍️ **Step 4:** Agent reasons about the bug and rewrites the code.
- 🔄 **Step 5:** Re-execution and verification of success.

| Risks of Full Autonomy



Infinite Loops & Resource Drain

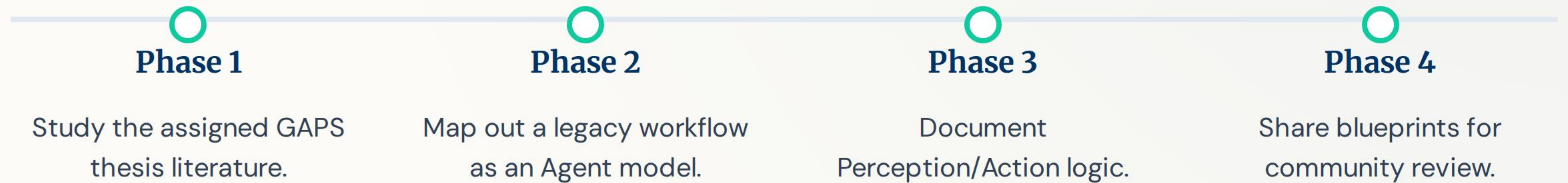
If not guarded, an agent can get stuck in a recursive loop, repeatedly failing and retrying, which consumes massive amounts of API tokens and computational resources without achieving the goal.

Human-in-the-Loop (HITL)

Safety & Governance for Agentic Systems

Speaker Notes: HITL is about balance. We give the agent autonomy, but we keep a human as the final authority for high-risk actions.

| Your Eid Holiday Action Plan



Speaker Notes: Use this break to move from theory to design. Choose a manual task and architect how an agent would solve it using the GAPS framework.

Questions?

Let's build the future of AI Development together.

www.panaversity.com | Student Portal

Image Sources



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